

SYNOPSIS

AI-BASED RESUME ATS SCORING AND JOB RECOMMENDATION SYSTEM

*A Web-Based Application for Resume Analysis,
ATS Compatibility Scoring and Personalized Job Recommendations*

Project Type	Web-Based Software Project
Frontend	React.js, Vite, Tailwind CSS
Backend	Python, FastAPI
Database	PostgreSQL
AI / NLP	spaCy, scikit-learn, Sentence Transformers
Development Environment	Visual Studio Code
Deployment for Development	Localhost

1. Introduction

The AI-Based Resume ATS Scoring and Job Recommendation System is a web application designed to help job seekers evaluate and improve their resumes while discovering job opportunities that closely match their skills, qualifications and experience. The system allows users to upload a resume in PDF or DOCX format, extracts relevant information, analyzes the resume against selected job requirements, generates an ATS compatibility score, identifies missing keywords and skills, and recommends suitable jobs.

The project combines web development, database management, Natural Language Processing (NLP), text similarity and recommendation techniques. The application is intended to provide an understandable and practical resume analysis rather than claiming to reproduce the proprietary scoring process of any particular Applicant Tracking System (ATS).

2. Problem Statement

Many job applicants submit resumes without knowing whether their resumes are compatible with automated Applicant Tracking Systems or whether they contain the skills and keywords required for a particular job. Manually checking a resume against multiple job descriptions is time-consuming and difficult. In addition, applicants may apply to jobs that do not closely match their skills and experience.

Therefore, there is a need for an integrated system that can analyze resumes, estimate ATS compatibility, provide actionable improvement suggestions, and recommend relevant jobs based on resume content.

3. Aim of the Project

The main aim of the project is to develop a user-friendly web application that uses resume parsing, ATS-oriented analysis and NLP-based similarity techniques to help users improve their resumes and identify relevant job opportunities.

4. Objectives

- Provide user registration, login and profile management.
- Allow users to upload resumes in PDF and DOCX formats.
- Extract and process text from uploaded resumes.
- Identify important skills, keywords, education and experience information.
- Generate an ATS compatibility score based on defined analysis criteria.
- Identify missing or weak keywords by comparing a resume with a job description.
- Provide practical suggestions for improving resume content.
- Recommend jobs based on similarity between resume information and job requirements.
- Provide a dashboard for viewing resume status, ATS results and recommended jobs.
- Develop the complete system so it can be tested locally through a browser.

5. Scope of the Project

The project will cover the following major areas:

- Resume upload and document processing.
- Resume text extraction and cleaning.
- Skill and keyword extraction.
- ATS compatibility analysis.
- Job description and resume comparison.
- NLP-based job matching and recommendation.
- User authentication and profile management.
- Job listing and recommendation display.
- Responsive frontend interface.
- REST API based communication between frontend and backend.

6. Proposed System

The proposed system consists of a React-based frontend, a Python FastAPI backend, a PostgreSQL database and an AI/NLP processing layer. The frontend provides the user interface, while the backend manages authentication, files, database operations and communication with the ATS and recommendation modules.

Overall workflow:

1. User registers or logs in.
2. User uploads a resume.
3. Backend validates and processes the document.
4. Resume text is extracted and cleaned.
5. Skills and keywords are identified.
6. The ATS module calculates an estimated compatibility score.
7. The system identifies missing keywords and provides suggestions.
8. The recommendation module compares resume information with available jobs.
9. Matching jobs are ranked and displayed to the user.

7. Major Modules

7.1 User Authentication and Profile Module

- User registration and login.
- Secure password hashing.
- JWT-based authentication.
- Protected API routes.
- User profile creation and updating.

7.2 Resume Management Module

- PDF and DOCX upload.
- File validation and storage.
- Resume text extraction.
- Resume record management.
- Viewing and deleting uploaded resumes.

7.3 ATS Analysis Module

- Keyword matching.
- Skill matching.
- Experience analysis.
- Education analysis.
- Basic resume formatting checks.
- Weighted ATS compatibility score.
- Missing keyword identification.
- Resume improvement suggestions.

7.4 Job Recommendation Module

- Maintain a structured job dataset.
- Extract relevant job skills and requirements.
- Represent resume and job text using TF-IDF or sentence embeddings.
- Calculate similarity between resume and jobs.
- Rank jobs by match percentage.
- Display recommended jobs to the user.

7.5 Dashboard Module

- Overall ATS score.
- Skill and keyword analysis.
- Missing keywords.
- Improvement suggestions.
- Recommended jobs.
- Resume and profile status.

8. ATS Scoring Approach

The system will use a defined weighted scoring model to produce an estimated ATS compatibility score. An initial model may assign weights to keyword matching, skills, experience, education and formatting. The weights can be adjusted after testing.

Component	Initial Weight
Keyword Match	30%
Skills Match	30%
Experience	20%
Education	10%
Formatting	10%
Total	100%

These weights are a proposed project model and are not an official score used by external ATS vendors. The final scoring model will be validated using sample resumes and job descriptions.

9. Job Recommendation Method

The recommendation engine will compare resume content with job requirements. Relevant skills and text features will be extracted from both sources. Similarity can be calculated using TF-IDF with cosine similarity for

a baseline model and Sentence Transformers for semantic matching. Jobs will then be sorted according to their calculated match score.

10. Technology Stack

Area	Technology
Frontend	React.js, Vite, Tailwind CSS, React Router, Axios
Backend	Python 3.11+, FastAPI, Uvicorn, Pydantic
Database	PostgreSQL, SQLAlchemy
Resume Processing	pdfplumber, python-docx
NLP / AI	spaCy, scikit-learn, Sentence Transformers, NumPy, Pandas
Testing	Postman, browser testing, Python tests
Development	Visual Studio Code

11. System Architecture

The system follows a client-server architecture. The React frontend communicates with FastAPI through REST APIs. FastAPI handles authentication, resume management, database operations and communication with the ATS and recommendation services. PostgreSQL stores users, profiles, resumes, jobs, ATS reports and recommendation data.

Architecture flow: User → React Frontend → FastAPI Backend → ATS/Recommendation Services → PostgreSQL Database → FastAPI → React Frontend.

12. Database Design

- Users: user identity and authentication information.
- Profiles: personal, education and professional profile details.
- Resumes: uploaded file metadata and extracted resume text.
- Skills: normalized skills associated with resumes or jobs.
- Jobs: job title, company, location, description, skills and other job information.
- ATS Reports: score, component scores, missing keywords and suggestions.
- Recommendations: user, job and calculated match percentage.

13. Team Work Distribution

Team Member	Role	Main Responsibilities
Person 1	Full Stack Developer	React frontend, responsive UI, API integration, dashboard, final integration and localhost execution.
Person 2	Backend Developer	FastAPI foundation, PostgreSQL, authentication, user/profile APIs, resume upload and core backend services.
Person 3	Backend / AI Developer	Resume NLP, skill extraction, ATS scoring, missing keywords, similarity model and job recommendation APIs.

14. Development and Execution Plan

10. Environment and software setup.
11. Frontend and backend project initialization.
12. Database and authentication implementation.
13. Resume upload and text extraction.
14. ATS scoring engine development.
15. Job recommendation engine development.
16. Frontend-backend API integration.
17. Dashboard and report visualization.
18. Testing, debugging and performance improvement.
19. Final localhost demonstration and documentation.

15. Hardware and Software Requirements

15.1 Recommended Hardware

- Laptop/Desktop with at least 8 GB RAM; 16 GB is recommended for smoother NLP/AI development.
- Modern multi-core processor.
- At least 10–20 GB free storage for development tools, dependencies and test files.
- Stable internet connection for installing packages and downloading NLP models.

15.2 Software Requirements

- Visual Studio Code.
- Python 3.11 or newer.
- Node.js LTS and npm (required on the full-stack developer's machine).
- PostgreSQL and pgAdmin.
- Postman.
- Google Chrome or another modern browser.

16. Expected Output

- A functional web application for resume analysis.
- Estimated ATS compatibility score out of 100.
- Skill and keyword analysis.
- Missing keyword identification.
- Actionable resume improvement suggestions.
- Personalized job recommendations with match percentages.
- User dashboard and profile management.
- Complete project running locally through localhost.

17. Advantages

- Saves time during resume screening and job searching.
- Helps users understand important resume weaknesses.

- Provides job recommendations based on user-specific information.
- Combines resume analysis and job matching in one platform.
- Can be extended with additional AI-based career features.

18. Limitations

- ATS scoring is an estimated compatibility score and may differ from proprietary ATS systems.
- Resume parsing accuracy may vary with unusual document layouts.
- Job recommendation quality depends on the quality and freshness of the job dataset.
- AI/NLP models may require tuning for different industries and job roles.

19. Future Scope

- Live job data integration through authorized APIs.
- AI-based resume rewriting.
- AI cover letter generation.
- Interview question generation based on the selected job.
- Career path recommendations.
- Multi-language resume analysis.
- Advanced semantic job matching.
- Cloud deployment and scalable infrastructure.
- Analytics for resume performance and application tracking.

20. Conclusion

The AI-Based Resume ATS Scoring and Job Recommendation System provides an integrated solution for two common challenges faced by job seekers: improving resume compatibility and finding relevant job opportunities. By combining React, FastAPI, PostgreSQL, NLP and machine-learning-based similarity techniques, the proposed system can analyze resumes, generate an estimated ATS score, identify missing keywords and recommend suitable jobs. The modular architecture also allows the system to be developed by a three-person team and tested completely on a local machine before deployment.

21. Keywords

Resume Analysis, Applicant Tracking System (ATS), ATS Score, Job Recommendation, Natural Language Processing, Machine Learning, FastAPI, React.js, PostgreSQL, Resume Parsing, Skill Extraction, Cosine Similarity, Sentence Embeddings.